Worked Examples: Logic Chains

Translating ecosystem accounts into policy-relevant values

EXAMPLE 1 Carbon Sequestration (Mangroves)

1	2	3	4	5
Mangrove extent	NCP rate	Physical supply	Social cost of carbon	Monetary value
350 ha	**3.7 Mg CO2/ha/yr**	**1,295 Mg CO2/yr**	**$51 / tCO2**	**$66,045 / yr**

350 ha x 3.7 Mg CO2/ha/yr = 1,295 Mg CO2/yr x $51/tCO2 = $66,045/yr

EXAMPLE 2 Fish Provisioning (Reef Fisheries)

1	2	3	4	5
Annual catch	Market price	Gross revenue	Minus costs	Resource rent
180,000 kg	**$3.50 / kg**	**$630,000**	**- $420,000**	**$210,000 / yr**

180,000 kg x $3.50/kg = $630,000 - $420,000 costs = $210,000/yr resource rent

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EXAMPLE 3 Coastal Protection (Coral Reefs)

1	2	3	4	5
Reef extent along coast	Assets behind reef	Replacement cost	Total replacement	Annualized value
12 km	**Infrastructure at risk**	**$5,000 / m**	**$60,000,000**	**$1,200,000 / yr**

12 km x 1,000 m/km x $5,000/m = $60M total, annualized at 2% = $1.2M/yr

How the methods connect

Carbon (Example 1)

Uses **extent account** (area) multiplied by literature rate, then valued using Social Cost of Carbon. Most accessible method -- needs only extent data and a published NCP rate.

Fish (Example 2)

Uses **catch statistics** and market data. Valued via resource rent (revenue minus intermediate costs). Requires economic data on fishing costs.

Coastal Protection (Example 3)

Uses **extent account** (reef length) and replacement cost of built infrastructure. Annualized to produce annual service flow.

Worked Examples: Logic Chains

Combined ecosystem services for one accounting area

EXAMPLE 4 Combined Services -- Single Accounting Area

#	Ecosystem Service	Ecosystem	Method	Physical Supply	Monetary Value ($/yr)
1	Fish provisioning	Coral reef	Resource rent	180,000 kg/yr	$210,000
2	Carbon sequestration	Mangrove + seagrass	Social cost of carbon	2,334 Mg CO2/yr	$119,085
3	Recreation / tourism	Coral reef + beach	Direct expenditure	25,500 visits/yr	$1,275,000
4	Coastal protection	Coral reef	Replacement cost	12 km reef frontage	$1,200,000
	TOTAL ANNUAL ECOSYSTEM SERVICE VALUE				$2,804,085

SHARE OF TOTAL VALUE

Fish Carbon Recreation Coastal protection

Key Insight: Regulating and cultural services dominate

Recreation (46%) and coastal protection (43%) together account for 89% of the total annual value. These non-market services are invisible without ocean accounts. Provisioning (fish) contributes only 7% of total value, yet it typically receives the most policy attention. Ocean accounts reveal the hidden economic importance of reef and mangrove conservation for tourism revenue and disaster risk reduction -- a powerful argument for Marine Spatial Planning and blue finance.